Issac Roy

Curriculum Vitae | San Diego, CA

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Research Interests

Embodied AI, robotics, and ML systems: efficient inference, 3D perception

Education

University of California San Diego **March 2027**

B.S. Data Science | *Math Minor* | *Cognitive Science Minor* | *GPA: 3.97/4.00* *San Diego, CA*

- **ML & Vision:** Computer Vision, Neural Networks and Deep Learning, Machine Learning, Probabilistic Modeling & ML
- **Math:** Honors Linear Algebra, Honors Multivariable Calculus, Honors Vector Calculus, Differential Equations, Probability, Statistics, Optimization for Data Science
- **Computer Science:** Data Structures and Algorithms, Theoretical Foundations of Data Science II, Systems for Scalable Analytics, Data Science Capstone, IoT and Sensors

Research Experience

Z-Lab | *Undergraduate Researcher (Capstone), 10 hrs/week* | *Prof. Zhijian Liu* **September 2026 – Present**

- Studying efficient inference for large models, including speculative decoding (DFlash) and quantization (AWQ, GPTQ).
- Planned project: accelerating $\pi_{0.5}$, Physical Intelligence's vision-language-action robot policy, on consumer GPUs.

MC Computer Vision Lab | *Undergraduate Researcher* | *Prof. Manmohan Chandraker* **January 2026 – May 2026**

- Designed a generative pipeline to turn coarse meshes into high-fidelity 3D assets with PBR materials, and built its mesh-generation stage on the National Research Platform's Kubernetes GPU cluster.
- Contributed to OpenRooms++, a photorealistic indoor-scene dataset for inverse rendering, surveying recent methods in generative 3D reconstruction (TRELLIS 2) and material estimation.

Independent Research | *GANs for Image Synthesis* | Technical Report **December 2025**

- Trained a DCGAN on Oxford Flowers 102 in PyTorch Lightning, stabilizing training with label smoothing, asymmetric learning rates and adaptive discriminator skipping; the production model reached FID 50.3.
- Ran 5 ablations of these stabilizers, scoring each with FID/KID/IS on held-out test images.
- Deployed the generator in Neural Nector, a full-stack web game where 100+ users tell real flowers from generated ones.

Industry Experience

Amazon Web Services | *Software Development Engineer Intern* | *EBS Snapshots* **September 2026 – Present**

- Built and deployed an automated pipeline on AWS (CDK, Fargate, EventBridge, DynamoDB, S3) that keeps on-call agents' knowledge bases in sync with their sources, such as code, team docs and PDF/DOCX files.
- Designed an extensible connector framework with a shared test suite, developed a change-significance classifier that blocks cosmetic changes, and gated every knowledge update behind regression tests.

Apple | *Software Engineer Intern* | *WebKit* **June 2026 – September 2026**

- Implemented `flex-wrap:balance`, a new CSS feature, in WebKit, the C++ engine behind Safari, using dynamic programming to minimize line-length imbalance, and built an interactive visualizer to compare layouts.
- Developed tooling that uses WebKit's test history to separate flaky failures from real regressions, clearing false positives from pre-commit testing and unblocking 50+ PRs per week.

Neurocrine Biosciences | *Data Engineer Intern* | *Clinical Solutions* **June 2025 – September 2025**

- Saved $100k annually by architecting a serverless containerized data ingestion pipeline on AWS (ECS, Lambda, S3, Redshift), eliminating dependency on third-party vendors.
- Saved 50+ hours / month for study teams by developing a data pipeline with LLM endpoints from AWS Bedrock to convert raw patient data into study narratives.

Projects

Unified Tag Links | *TypeScript, Obsidian API, Vitest* **October 2026 – Present**

- Built and published an open-source plugin, downloaded 200+ times, in Obsidian, a note-taking app with 1.5M+ users.
- Designed a concept-note index that resolves nested tags to their deepest matching note, keeping a managed frontmatter property in sync as notes change, tested with Vitest and GitHub Actions CI.

Project Daedalus (G-Class Rocket) | *C++, KiCad, Onshape, Arduino* **October 2025 – Present**

- Wrote real-time flight software in C++ on an Arduino Nano Every, reading an IMU and barometer over I2C/SPI, with LoRa telemetry and black-box logging on a custom PCB.
- Designed & 3D-printed avionics bay to sustain high-G loads and protect electronics, targeting a 5,500 ft (1 mi+) apogee.

OnlyDance | *Python, MediaPipe, OpenCV, FastAPI, React Three Fiber* **April 2025**

- Won 1st place at Datahacks 2025 by developing an interactive full-stack app that teaches dance routines using 3D avatars and real-time feedback.
- Devised a real-time similarity scoring algorithm using MediaPipe to extract 3D pose landmarks, implementing sliding window techniques and frame interpolation to quantify dance accuracy.

Turbo 3D Engine | *C++, SDL2, Linear Algebra* **March 2025**

- Built a 3D graphics engine from scratch, implementing a custom rasterization pipeline, matrix transformations for camera space, custom material system, texture mapping, and multiple lighting models without external graphics libraries.
- Implemented multi-threaded rendering to parallelize vertex processing and rasterization across CPU cores, achieving playable frame rates through iterative performance profiling and optimization.
- Debugged rendering artifacts and memory corruption using GDB; isolated race conditions in the multi-threaded pipeline and fixed synchronization issues.

Self-Playing Guitar | *C++, Arduino, Laser Cutting, 3D Printing* **October 2023**

- Engineered a self-playing guitar with custom laser-cut and 3D-printed parts, programming a note transcriber algorithm to map musical inputs to precise servo motions.

Leadership & Teaching

AWS Student Builder Group at UCSD | *Founder & Technical Lead* **November 2025 – Present**

- Founded AWS SBG to teach students cloud computing, serverless architecture, and industry best practices.
- Leading technical workshops on core AWS services, equipping 200+ members with practical cloud skills.

Data Science Student Society (DS3) | *Software Lead* **October 2024 – June 2026**

- Led a 10-person team to build a membership portal for 1,000+ students. Established automated CI/CD pipelines via GitHub Actions and introduced Agile methodologies to improve productivity and accountability.
- Developed the website frontend in React and engineered a Supabase backend, incorporating auth and schemas for member tracking, event attendance, and analytics.

theCoderSchool | *Code Coach* **June 2024 – October 2024**

- Led 4 week-long summer coding camps, teaching Python and Lua to K-12 students. Remained undefeated in foosball.

Mathnasium | *Learning Instructor* **July 2022 – June 2024**

- Tutored K-12 students in math, teaching for intuition over memorized procedures so concepts carry to new problems.

Awards

1st Place, DataHacks | *OnlyDance, 50+ teams* **April 2025**

3rd Place, DataHacks | *Dino Classifier, 30+ teams* **March 2024**

1st Place, IEEE Fall Competition | *Self-Playing Guitar, 30+ teams* **October 2023**

Provost Honors, UC San Diego | *9 of 9 Quarters* **Fall 2023 – Present**

Technical Skills

Languages: C++, C, Python, Java, SQL, TypeScript, JavaScript, Bash, HTML/CSS
Tools: AWS, PyTorch, React, Terraform, Docker, Git, CI/CD, Postgres, Linux, Raspberry Pi, Arduino, KiCad, Onshape
Certifications: AWS AI Practitioner, AWS Cloud Practitioner, AWS Solutions Architect – Associate (in progress)